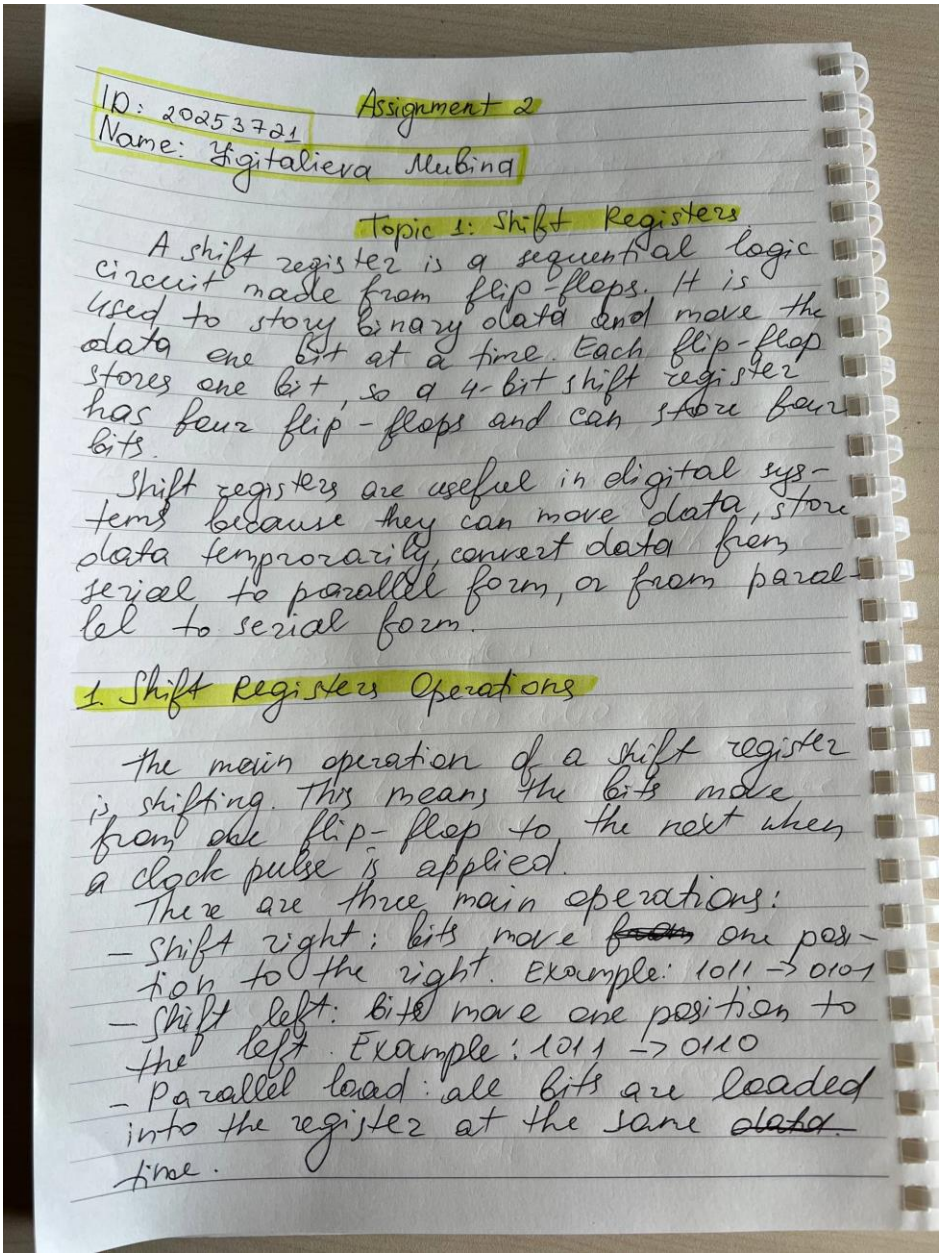
Assignment 2 – Logic Circuit

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Assignment 2

Topic 1: Shift Registers

A shift register is a sequential logic circuit made from flip-flops. It is used to store binary data and move the data one bit at a time. Each flip-flop stores one bit, so a 4-bit shift register has four flip-flops and can store four bits.

Shift registers are useful in digital systems because they can move data, store data temporarily, convert data from serial to parallel form, or from parallel to serial form.

1. Shift Registers Operations

the main operation of a shift register is shifting. This means the bits move from one flip-flop to the next when a clock pulse is applied.

There are three main operations:

- Shift right: bits move one position to the right. Example: 1011 -> 0101
- Shift left: bits move one position to the left. Example: 1011 -> 0110
- Parallel load: all bits are loaded into the register at the same time.

The clock signal controls the movement of data. When there is no clock pulse, the data stays in the register.

2. Types of Shift Register Data I/Os

Shift registers have different input and output methods.

SISO: Serial-In Serial-OUT
Data enters one bit at a time and leaves one bit at a time. It is useful for delaying data.

SIPO: Serial-In Parallel-Out
Data enters one bit at a time, but the output is taken all together. It converts serial data into parallel data.

PISO: Parallel-In Serial-Out
Data is loaded all at once but it leaves one bit at a time; it converts parallel data into serial data.

PIPO: Parallel-In Parallel-Out
Data enters all at once and leaves all at once. It is mostly used for temporary storage.

3. Bidirectional shift registers

A bidirectional shift register can shift data in both directions: left and right. It has a control input to choose the direction.

For example: if control = 0, data shifts right. If control = 1, data shifts left.

This type is useful because it is more flexible than a normal shift register. It is used in data transfer, arithmetic operations, and digital systems where data must move in both directions.

4. Shift Register counters

A shift register can also work as a counter. In this case the output is connected back to the input to make a repeating sequence.

- Ring Counter

In a ring counter, the last output is connected back to the first input.
Example for 4 bits:

1000 → 0100 → 0010 → 0001 → 1000

Only one bit is 1 at a time.

- Johnson Counter

A Johnson counter is also called a twisted ring counter, the inverted output of the last flip-flop is connected to the first input. Example 4 bits:

0000 → 1000 → 1100 → 1110 → 1111 → 0111 →
→ 0011 → 0001 → 0000

It gives more states than a ring counter.

Conclusion:

Shift registers are important in logic circuits because they can store and move binary data. They can shift data left or right, load data in parallel, and convert data between serial and parallel forms. They can also be used as counters, like ring counters and Johnson counters.